QPCODE: **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) DEGREE EXAMINATION
FOURTH SEMESTER

(2024 ADMISSION ONWARDS)

Discipline Specific Core Course (DSC C)

MG4DSCCHE202: FUNDAMENTALS OF PHYSICAL CHEMISTRY

Duration: 1.5 Hr **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A – Multiple Choice Questions (1×10 = 10 marks)

(Answer all questions. Each question carries 1 mark).

1. A face-centered cubic (FCC) unit cell contains atoms at corners and face centers. Which feature mainly contributes to the higher packing efficiency of FCC structures? [CO 1] [U]

a) Larger atomic radius b) Presence of atoms only at corners

c) Close packing due to face-centered atoms d) Increased unit cell volume

2. The adsorption of gases on solids is generally; [CO 2] [U]

a) Endothermic b) Exothermic

c) Neutral d) None

3. How many atoms are effectively present in a simple cubic (SC) unit cell? [CO 1] [U]

a) 1 b) 2

c) 4 d) 6

4. Lyophilic colloids are [CO 3] [U]

a) Easily coagulated b) Difficult to coagulate

c) Insoluble d) Unstable

5. In ultrafiltration, what would most likely happen if the pore size of the membrane is increased beyond the size of colloidal particles?? [CO 3] [An]

a) Purification efficiency increases b) Only ions will pass through

c) Colloidal particles will also pass through, reducing separation efficiency

d) Pressure requirement decreases without affecting purity

6. When NaCl is added to colloidal solution of Fe (OH)3 the colloid gets coagulated. This is because : [CO 3] [A]

a) Colloidal particles gain same charge as Na2 ions b) pH of solution decreases

c) Cl2 ions neutralize the positive charge on Fe(OH)3 d) Na2 ions increases Brownian motion

7 Potentiometric titration is based on the measurement of [CO 4] [U]

a) pH b) EMF

c) Conductivity d) Temperature

8. What is the standard electrode potential of Hydrogen electrode? [CO 4] [k]

a) 0 b) 1 V

c) -1V d)+1 V

9. If 0.916gm of silver is deposited at a cathode by passing 0.1A current for 1000sec. the equivalent weight of silver is [CO 4] [A]

a) 107.9 b) 96.5

c) 10.79 d) 1079

10. The Nernst equation is mainly used to determine [CO 4 [U]

a) Atomic mass of an element b) Electrode potential at non-standard conditions

c) Rate of electrochemical reaction d) Electrical conductivity of solution

Part B – Short Answer Questions (4×5 = 20 marks)

(Answer any **Four** questions. Each question carries **5** marks.)

11. Distinguish between Physisorption and Chemisorption [CO 2] [U]

12. Analyze the role of zeta potential in the stability of colloids. How does coagulation occur according to the Hardy–Schulze rule? [CO3] [An]

13. Explain how molar conductance varies with dilution for strong and weak electrolytes. Give reasons for the observed trends. [CO 4] [U]

14. Using Kohlrausch's law of independent migration of ions, explain how the limiting molar conductance of a weak electrolyte can be determined. [CO 4] [A]

15. Explain the band theory of solids and use it to differentiate between conductors, semiconductors, and insulators. [CO 1] [U]

16. Explain Brownian movement and Tyndall effect. How do these properties help in identifying colloidal systems?. [CO 3] [U]

Part C – Short Essay Questions (2×10 = 20 marks)

(Answer any **two** questions. Each question carries **10** marks.)

17. Derive the Nernst equation for a galvanic cell and explain how it is used to calculate electrode potential under non-standard conditions. [CO 4] [A]

18. Explain in detail the electrical and magnetic properties of solids [CO 1] [U]

19. Apply the concept of colloids to explain their role in medicines and delta formation. Discuss the advantages of colloidal systems in drug delivery [CO 3] [A]

20. Explain different types of indicator electrodes with suitable examples: metal–metal ion electrode, quinhydrone electrode, and redox electrode. [CO4] [U]